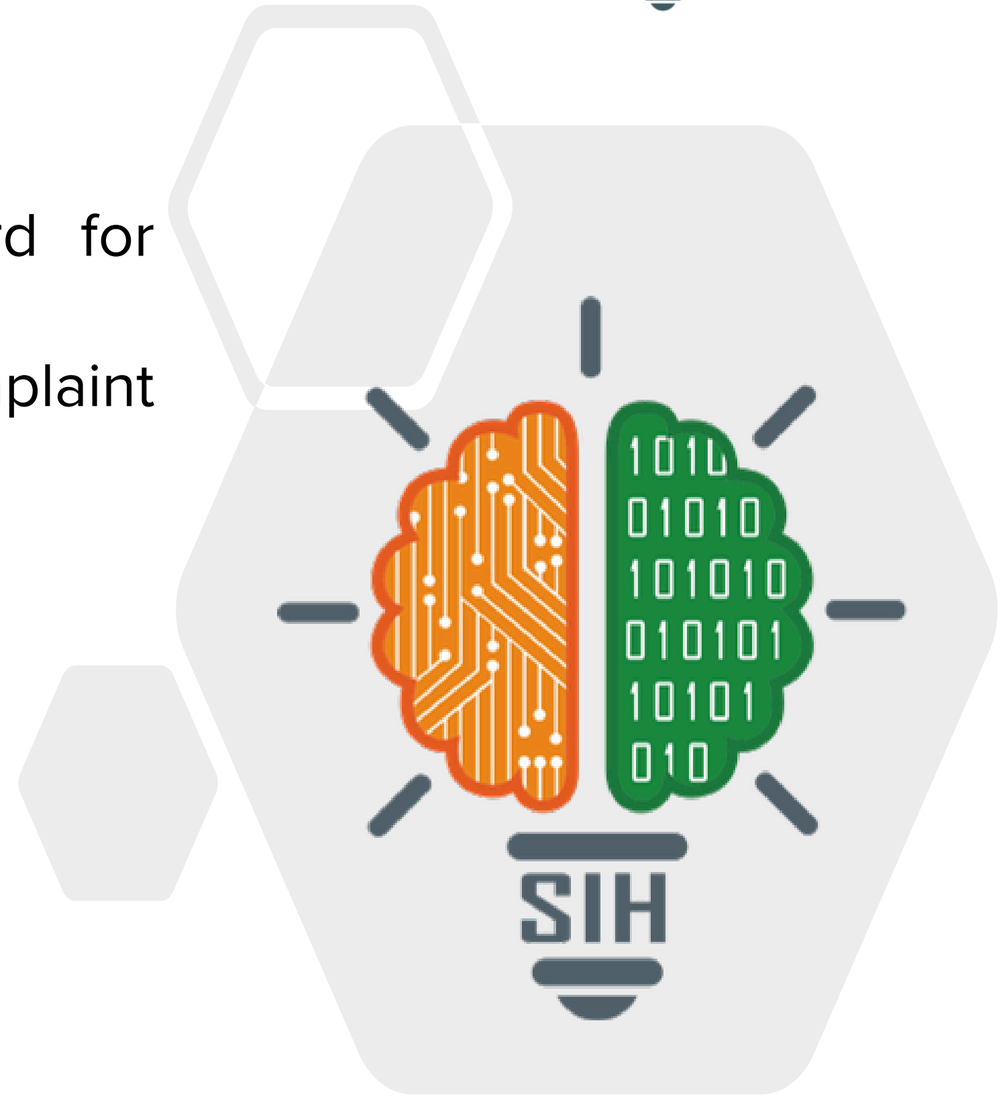
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- **Problem Statement ID –** 26264
- **Problem Statement Title-** Integrated Dashboard for Monitoring and Tracking Institution-Related Complaint Cases Across AICTE Bureaus/Cells
- **Theme-** Smart Automation
- **PS Category-** Software
- **Team ID-**
- **Team Name -** Gremlinix



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ResolveGrid — Intelligent Complaint Monitoring & Coordination

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One Institution. Connected Cases. Coordinated Action.

Addressing the problem

AICTE receives complaints related to technical education institutions from students, faculty members and other stakeholders.

Different Bureaus/Cells, including RB and PGRC, may handle complaints concerning the same institution, creating a risk of overlapping actions and duplicated efforts.

Lack of a centralized view makes it difficult to track case status, identify responsible departments and monitor deadlines.

Need: A centralized complaint intelligence platform that improves cross-bureau coordination, case tracking and timely resolution.

Uniqueness and Innovation

- Centralized Institution-Wise Complaint History
- AI-Assisted Cross-Bureau Overlap Detection
- Explainable Alerts for Potentially Related Cases
- Smart Deadline Monitoring and Escalation
- Real-Time Case Analytics and Workload Tracking



ResolveGrid

ResolveGrid is a centralized web-based complaint monitoring platform that helps authorized AICTE officials track institution-related cases, identify potential overlaps and coordinate actions across Bureaus/Cells.

PROPOSED SOLUTION

ResolveGrid
Complaint Intelligence Platform

COMPLAINT REGISTRATION
Register complaints and capture institution and case details

INSTITUTION PROFILE
View consolidated complaint history for each institution.

CROSS-BUREAU MATCHING
Identify potentially related cases using institution details and complaint similarity.

OVERLAP ALERTS
Notify officials when related cases require coordination.

CASE TRACKING & ESCALATION
Monitor status, assigned officers, deadlines and action history.

SMART DASHBOARD
Visualize pending cases, overdue cases, department workload and disposal trends.

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TECHNICAL APPROACH

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TECH STACK

- React.js + Tailwind CSS
- Python + FastAPI
- PostgreSQL
- Pandas
- Scikit-learn
- Rule-Based Engine
- Git + GitHub

SOLUTION ARCHITECTURE



USER REGISTRATION & LOGIN

Authorized AICTE officials sign in according to their roles and permissions.

COMPLAINT REGISTRATION

Capture Complaint ID, Institution ID, category and description.

CENTRAL SERVICE API

FastAPI processes requests, validates data and manages complaint workflows.

INSTITUTION & CASE IDENTIFICATION

Match institution records and retrieve related complaint histories.

OVERLAP DETECTION ENGINE

Compare institution IDs, complaint categories and text similarity to flag potentially related cases.

DATABASE

PostgreSQL stores institution profiles, complaints, assignments, action history and notifications.

COORDINATION & ALERT ENGINE

Generate potential-overlap alerts, pending-case reminders and deadline-based escalations.

CASE TRACKING & WORKFLOW

Support case assignment, status updates, transfers, remarks and action-history tracking.

ANALYTICS DASHBOARD

Display pending cases, overdue cases, Bureau/Cell workloads, potential overlaps and disposal trends.

REPORTING & MONITORING

Display pending cases, overdue cases, Bureau/Cell workloads, potential overlaps and disposal trends.

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FEASIBILITY AND VIABILITY

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Technical Feasibility

- Open-source technologies
- Modular and scalable architecture
- Lightweight AI-based complaint matching



Operational Feasibility

- Simple mobile-based interface
- Role-based access for authorized officials
- Complaint Tracking → Overlap Detection → Alerts → Resolution Monitoring



Market Feasibility

- Supports AICTE Bureaus/Cells
- Addresses cross-department coordination gaps
- Adaptable to other educational regulatory bodies



Viability

Easy user adoption
Simple interface for officials

Scalable solution
Department-level pilot → Organization-wide deployment

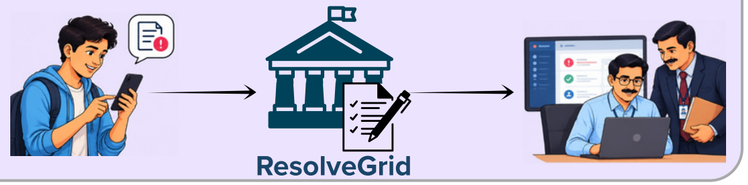
Sustainable model
Supports Farmers, FPOs and Buyers in one ecosystem

Economic Viability

Stage	Indicative Cost
SIH Prototype	₹0 target
Limited Pilot	₹5–15 Lakh
Organization-Wide Deployment	₹25–60+ Lakh

Indicative planning estimates only

Actual implementation costs depend on user volume, integrations, hosting, cybersecurity, data migration, maintenance and procurement requirements.



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IMPACT AND BENEFITS

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IMPACTS

ECONOMIC IMPACTS

- Reduced duplication of administrative work
- Better utilization of staff and departmental resources
- Reduced time spent identifying overlapping cases
- Improved efficiency in complaint processing

ADMINISTRATIVE IMPACTS

- Improved coordination across AICTE Bureaus/Cells
- Reduced risk of conflicting or parallel actions
- Faster identification of overdue complaints
- Better visibility of institution-related cases

TRANSPARENCY AND ACCOUNTABILITY

- Complete chronological case history
- Clear identification of responsible Bureau/Cell
- Traceable case transfers and escalations
- Improved monitoring of complaint status

BENEFITS

AICTE OFFICIALS

- Centralized complaint monitoring
- Easier case tracking and assignment
- Timely alerts for pending and overdue cases
- Better workload management

REGULATION BUREAU (RB) & PGRC

- Visibility of potentially overlapping cases
- Easier coordination between departments
- Consolidated institution-wise complaint history
- Reduced repetitive effort

EDUCATIONAL INSTITUTIONS

- Potentially more coordinated complaint handling
- Improved visibility of case progress, where authorized
- Reduced risk of duplicate administrative processing

AICTE MANAGEMENT

- Bureau-wise and region-wise analytics
- Case ageing and disposal reports
- Better oversight of pending complaints
- Data-informed administrative decisions

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RESEARCH AND REFERENCES

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"Research Behind Our Solution"

PRIMARY SYSTEM REFERENCE

AICTE — Grievance Redressal Regulations

Website: India Code — AICTE Regulations, 2012
Studied institutional grievance mechanisms, complaint categories, and procedures for complaint handling.

Inspired our solution: Centralized case records, status tracking, and deadline monitoring.

AICTE — Approval Process Handbook

Website: Official AICTE PDF
Reviewed the handbook's references to grievance redressal arrangements for students and faculty.

Inspired our solution: Coordinated monitoring and clearer institutional accountability.

RESEARCH

2017

Similarity-Based Approach for Online Complaints

Authors: Xiaolan Cui, Shuqin Cai and Yuchu Qin.
Research on retrieving similar complaint cases using word-level and text-level similarity.
Link: DOI: 10.1108/K-10-2016-0271

2021

Service Failure and Complaints Management in Higher Education Institutions

Author: Steven Kayambazinthu Msosa.
Study of complaint management and service failures across three public universities in South Africa.
Link: Read the research paper

2026

AI-Driven Complaint Management System

Research on complaint prioritization, tracking, escalation, and duplicate detection.
Link: IEEE conference paper

"Previously reported complaint issues"

